

both rule-based and machine-learning-driven approaches for natural conversations.

Use cases:

- Automates customer support by answering FAQs.
- Provides personalised recommendations based on user queries.
- Enhances business communication with real-time chat assistance.

Basic Rasa chatbot code:

```
# Install Rasa
pip install rasa
# Initialize Rasa project
rasa init
# Train the chatbot model
rasa train
# Run the chatbot
rasa shell
```

Speech-to-text converter with OpenAI Whisper

Whisper is an advanced speech recognition model by OpenAI that converts spoken language into text. It is useful for transcription services, voice assistants, and accessibility tools.

Use cases:

- Converts interviews or meetings into text automatically.
- Provides real-time subtitles for videos.
- Helps in accessibility solutions for hearing-impaired individuals.

Code example:

```
import whisper
# Load the Whisper model
model = whisper.load_model("base")
# Transcribe an audio file
result = model.transcribe("audio.mp3")
# Print the text output
print(result["text"])
```

Resume screening AI

Resume screening AI uses natural language processing (NLP) to analyse job applications and match them to job descriptions. It ranks resumes based on skills, experience, and relevance.

Use cases:

- HR departments can automate candidate shortlisting.
- Reduces bias in the hiring process by focusing on qualifications.
- Saves time by filtering out irrelevant applications.

Code for resume screening using NLP:

```
import spacy
# Load the English NLP model
```

```
nlp = spacy.load("en_core_web_sm")
# Example resume text
resume_text = "Experienced software engineer skilled in Python, AI, and machine learning."
# Process the text
doc = nlp(resume_text)
# Extract key skills (example using named entities)
skills = [ent.text for ent in doc.ents]
print("Extracted Skills:", skills)
```

AI for medical diagnosis

AI is widely used in medical image analysis for detecting diseases such as cancer, pneumonia, and COVID-19. Deep learning models (CNNs, ResNet, U-Net, etc) are trained on medical imaging datasets.

Use cases:

- Helps radiologists detect abnormalities in X-rays and MRIs.
- Enhances early disease detection and diagnosis accuracy.
- Reduces workload in hospitals by automating analysis.

Example: Using TensorFlow for medical image classification

```
import tensorflow as tf
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing import image
import numpy as np
# Load a pre-trained model
model = load_model("medical_diagnosis_model.h5")
# Load and preprocess the image
img = image.load_img("chest_xray.jpg", target_size=(224, 224))
img_array = image.img_to_array(img) / 255.0
img_array = np.expand_dims(img_array, axis=0)
# Make a prediction
prediction = model.predict(img_array)
print("Diagnosis:", "Positive" if prediction[0] > 0.5 else "Negative")
```

AI-based language translation

MarianMT and OpenNMT are open source machine translation models that allow real-time language translation for various applications.

Use cases:

- Businesses use it for multilingual customer support.
- Travellers use AI-powered translators for communication.
- Websites provide automatic content localisation.

Example: Translate text using MarianMT (Hugging Face)

```
from transformers import MarianMTModel, MarianTokenizer
# Define source and target language (English to French)
```